# Md Abid Al Morshed | Computational Mechanics | Composite fracture

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## CORE COMPETENCIES

Mechanical Engineering PhD candidate with experience in **Abaqus FEA, structural simulation, and failure analysis**. Skilled in custom FEA implementation (**UEL/UMAT**), composite material modeling, and simulation-to-experiment comparison. Seeking an internship in **CAE, structural mechanics, or mechanical engineering R&D**.

## EXPERIENCE

**Graduate Research Assistant** | Fall 2024–Present **Michigan Technological University**

- Developed a 3D phase-field model for composite fracture under static and dynamic loading.
- Implemented the formulation through custom Fortran user elements in Abaqus.

**Graduate Teaching Assistant** | Spring 2025 **Michigan Technological University**

- Led practical sessions in control engineering and HVAC analysis.
- Taught MATLAB/Simulink and Python for engineering modeling and data analysis.

## EDUCATION

**Michigan Technological University | GPA: 3.84** Expected graduation: Fall 2027

Ph.D. in Mechanical and Aerospace Engineering | Computational modeling of composite fracture

**Bangladesh University of Engineering and Technology | GPA: 3.69** Graduated: May 2023

B.Sc. in Mechanical Engineering | Computational fluid mechanics and heat transfer

## ENGINEERING PROFICIENCIES

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| **Engineering analysis:** | Nonlinear FEA, structural failure analysis, fracture mechanics, composite failure. |
| **Software & Programming:** | Abaqus (custom user elements: UEL/UMAT), Fortran, Python, MATLAB, Simulink, Linux. |

## HONORS AND PROFESSIONAL INVOLVEMENTS

**Awards:** College of Engineering Candidacy Incentive Award, Michigan Tech (2026); Dean's List and University Merit Scholarship, BUET (2017, 2018).

**Conference Presentations:** American Society for Composites (2026); Society of Engineering Science (2026).

**Research Communication:** Three-Minute Thesis, Michigan Tech (2025, 2026), ICC computing showcase (2026).

**Mentoring & Community Service:** Research Rotation Mentor, COE–REU Program (2026); Festival of Nations (2026); Keweenaw Land Trust trail cleanup (2025).

## SELECTED PUBLICATONS

**Journal Paper**

- **MAA Morshed**, NA Mini, MA Hakim, and MN Hasan (2025). "Mixed convective heat transfer characteristic in vented cavity under dynamic flow modulation by CFD and neural network model approaches". *Heat Transfer, 54*(5), 3072–3087. DOI:10.1002/htj.23349
- **MAA Morshed** and T Sain. "Towards Dynamic Fracture Modeling of Fiber-Reinforced Composites Using a Mixed-Mode Energy Decomposition-Based Phase-Field Approach." *Engineering Fracture Mechanics*, under review, 2026.
- **MAA Morshed,** MR Karim, A Kumar and T Sain. "A Three-dimensional Mode-resolved Phase-field Formulation for Quasi-static and Dynamic Intralaminar Fracture in Unidirectional Composite." *Computer Methods in Applied Mechanics and Engineering*, under review, 2026.

**Book Chapter**

- **MAA Morshed** and NA Mini. "Lessons Learned from CFRP Failures: Case Studies and Insights." In Handbook of Carbon Fiber Reinforced Polymer Composites, edited by Md Zillur Rahman and Kalim Deshmukh. *Springer Nature*, (accepted), 2026.

**Conference Proceeding**

- **MAA Morshed**, MA Mahmud, M Atiqur, and MZ Haq. "Thermal Analysis of an Advanced Adiabatic Compressed Air Energy Storage System with Phase Changing Materials for Thermal Energy Storage." *Proceedings of the Energy Conference 2023: National and Global Issues (ENCON23)*, 2024. DOI:10.2139/ssrn.4993585
- **MAA Morshed** and T Sain. "Predicting Fracture in Fiber-Reinforced Composites Using a Novel Mixed-Mode Energy Decomposition Based Phase-Field Model Under Dynamic Loading." *American Society for Composites 41st Annual Technical Conference*, (accepted), 2026.